References

[1] 2024. Wiley's 'fake science' scandal is just the latest chapter in a broader crisis of trust. *ABC News (Australia)* (May 2024). Published May 20, 2024.

[2] 2026. HalluCitation Matters: Revealing the Impact of Hallucinated Citations. *arXiv* (Jan. 2026). arXiv:2601.18724.

[3] Ibrahim Adabara, Bashir Olaniyi Sadiq, Aliyu Nuhu Shuaibu, Yale Ibrahim Danjuma, and Venkateswarlu Maninti. 2025. Trustworthy Agentic AI Systems: A Cross-Layer Review of Architectures, Threat Models, and Governance Strategies for Real-World Deployment. *F1000Research* 14 (2025), 905. doi:10.12688/f1000research.169927.1

[4] Gastão Amaral et al. 2024. ProVe: A pipeline for automated provenance verification of knowledge graphs against textual sources. *Semantic Web Journal* (2024).

[5] Anthropic. 2025. Agentic Misalignment: How LLMs Could Be Insider Threats. https://www.anthropic.com/research/agentic-misalignment. Accessed: 2026-02-04.

[6] Rauno Arike, Elizabeth Donoway, Henning Bartsch, and Marius Hobbhahn. 2025. Technical Report: Evaluating Goal Drift in Language Model Agents. arXiv:2505.02709 [cs.AI] doi:10.48550/arXiv.2505.02709

[7] Adam Bates, Dave Tian, Kevin R. B. Butler, and Thomas Moyer. 2015. Trustworthy Whole-System Provenance for the Linux Kernel. In *24th USENIX Security Symposium (USENIX Security 15)*. USENIX Association, 319–334.

[8] Joeran Beel and Min-Yen Kan. 2025. Evaluating Sakana's AI Scientist: Bold Claims, Mixed Results, and a Promising Future? *arXiv preprint arXiv:2502.14297* (2025). https://arxiv.org/abs/2502.14297

[9] Khalid Belhajjame, Reza B'Far, James Cheney, Sam Coppens, Yolanda Gil, Paul Groth, Graham Klyne, et al. 2013. *PROV-DM: The PROV Data Model.* Technical Report. W3C Recommendation. https://www.w3.org/TR/prov-dm/

[10] Bernal Bernal, Weizhe Yuan, O Harari, et al. 2024. HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 37.

[11] Maciej Besta, Lorenzo Paleari, Jia Hao Andrea Jiang, Robert Gerstenberger, et al. 2025. Affordable AI Assistants with Knowledge Graph of Thoughts. *arXiv preprint arXiv:2504.02670* (2025).

[12] Siddhant Bhambri, Upasana Biswas, and Subbarao Kambhampati. 2025. Do Cognitively Interpretable Reasoning Traces Improve LLM Performance? *arXiv preprint arXiv:2508.16695* (2025). https://arxiv.org/abs/2508.16695

[13] Wulfram Blau, Vinton G. Cerf, Juan Enriquez, Joseph S. Francisco, Urs Gasser, Mary L. Gray, Mark Greaves, Barbara J. Grosz, Kathleen Hall Jamieson, Gerald H. Haug, et al. 2024. Protecting scientific integrity in an age of generative AI. *Proceedings of the National Academy of Sciences* 121, 22 (2024), e2407886121. doi:10.1073/pnas.2407886121

[14] Daniil A Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. 2023. Autonomous chemical research with large language models. *Nature* 624, 7992 (2023), 570–578. doi:10.1038/s41586-023-06792-0

[15] Vasilis Bountris, Lauritz Thamsen, and Ulf Leser. 2025. HyProv: Hybrid Provenance Management for Scientific Workflows. *arXiv preprint arXiv:2511.07574* (2025).

[16] Samuel R Bowman, Gabor Angeli, Christopher Potts, and Christopher D Manning. 2015. A large annotated corpus for learning natural language inference. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*. 632–642.

[17] Andres M Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D White, and Philippe Schwaller. 2024. ChemCrow: Augmenting large-language models with chemistry tools. *Nature Machine Intelligence* 6 (2024), 525–535. doi:10.1038/s42256-024-00832-8

[18] C Brundy. 2024. The paper mill crisis is a five-alarm fire for science. *Insights: the UKSG journal* (2024). Discusses Wiley/Hindawi retractions exceeding 11,300 as of April 2024.

[19] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, et al. 2025. Why Do Multi-Agent LLM Systems Fail? *arXiv preprint arXiv:2503.13657* (2025).

[20] Mikaël Chelli, Jules Descamps, Vincent Lavoué, Christophe Trojani, Michel Azar, Marcel Deckert, Jean-Luc Raynier, Gilles Clowez, Pascal Boileau, and Caroline Ruetsch-Chelli. 2024. Hallucination Rates and Reference Accuracy of ChatGPT and Bard for Systematic Reviews: Comparative Analysis. *J Med Internet Res* 26 (22 May 2024), e53164. doi:10.2196/53164

[21] Mikaël Chelli, Jules Descamps, Vincent Lavoué, Christophe Trojani, Michel Azar, Marcel Deckert, Jean-Luc Raynier, Gilles Clowez, Pascal Boileau, and Caroline Ruetsch-Chelli. 2024. Hallucination Rates and Reference Accuracy of ChatGPT and Bard for Systematic Reviews: Comparative Analysis. *Journal of Medical Internet Research* 26 (2024), e53164. doi:10.2196/53164

[22] I Chern, Steffi Chern, Shiqi Chen, Weizhe Yuan, K Feng, C Zhou, J He, Graham Neubig, and Pengfei Liu. 2024. FacTool: Factuality Detection in Generative AI – A Tool Augmented Framework. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*.

[23] Mingxuan Du, Benfeng Xu, Chiwei Zhu, Xiaorui Wang, and Zhendong Mao. 2025. DeepResearch Bench: A Comprehensive Benchmark for Deep Research Agents. *arXiv preprint arXiv:2506.11763* (2025).

[24] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2023. Improving Factuality and Reasoning in Language Models through Multiagent Debate. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36. 31114–31136.

[25] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. 2024. Detecting hallucinations in large language models using semantic entropy. *Nature* 630 (2024), 625–630. doi:10.1038/s41586-024-07421-0

[26] Paul Fernsel, Cedric Maier, Ehsan Toreini, and Dominik Kowald. 2024. Assessing the Auditability of AI-integrating Systems: A Framework and Learning Analytics Case Study. *arXiv preprint arXiv:2411.08906* (2024).

[27] Sharon Goldman. 2026. NeurIPS research papers contained 100+ AI-hallucinated citations. *Fortune* (Jan. 2026). Published Jan 21, 2026.

[28] Jutta Haider, Björn Ekström, Kristofer Söderström, and Malte Rödl. 2024. AI-fabricated "junk science" floods Google Scholar: Key features, spread, and implications for preempting evidence manipulation. https://www.hb.se/en/about-ub/current/news-archive/2024/december/ai-fabricated-junk-science-floods-google-scholar/

[29] Thomas Hartung. 2025. AI, agentic models and lab automation for scientific discovery. *Frontiers in Artificial Intelligence* (2025). Frontiers.

[30] Melanie Herschel, Ralf Diestelkämper, and Houssem Ben Lahmar. 2017. A survey on provenance: What for? What form? What from? *The VLDB Journal* 26, 6 (2017), 881–906.

[31] Yikuan Hu, Jie Zhou, Ganqu Cui, Shengding Hu, and Maosong Sun. 2025. RemindRAG: Low-Cost LLM-Guided Knowledge Graph Traversal for Efficient RAG. *arXiv preprint arXiv:2510.13193* (2025).

[32] Moritz Baumgart Joeran Beel, Min-Yen Kan. 2025. Evaluating Sakana's AI Scientist for Autonomous Research. *arXiv preprint arXiv:2502.14297* (2025). https://arxiv.org/html/2502.14297v2

[33] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling Laws for Neural Language Models. *arXiv preprint arXiv:2001.08361* (2020).

[34] Roham Koohestani et al. 2025. AgentGuard: Runtime Verification of AI Agents. *arXiv preprint arXiv:2509.23864* (2025).

[35] Minghao Li, Ying Zeng, Zhihao Cheng, Cong Ma, and Kai Jia. 2025. ReportBench: Evaluating Deep Research Agents via Academic Survey Tasks. *arXiv preprint arXiv:2508.15804* (2025). https://arxiv.org/abs/2508.15804

[36] Miles Q. Li, Benjamin C. M. Fung, Martin Weiss, Pulei Xiong, Khalil Al-Hussaeni, and Claude Fachkha. 2026. A Benchmark for Evaluating Outcome-Driven Constraint Violations in Autonomous AI Agents. arXiv:2512.20798 [cs.AI] https://arxiv.org/abs/2512.20798

[37] Q. Vera Liao and Jennifer Wortman Vaughan. 2024. AI Transparency in the Age of LLMs: A Human-Centered Research Roadmap. *Harvard Data Science Review* 6, 2 (2024).

[38] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2023. Let's Verify Step by Step. *arXiv preprint arXiv:2305.20050* (2023).

[39] Zachary C Lipton. 2018. The Mythos of Model Interpretability. *Queue* 16, 3 (2018), 31–57.

[40] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the Middle: How Language Models Use Long Contexts. *Transactions of the Association for Computational Linguistics* 12 (2024), 157–173.

[41] Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. 2024. The AI Scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292* (2024). https://arxiv.org/abs/2408.06292

[42] Ruofan Lu, Yichen Li, and Yintong Huo. 2025. Exploring Autonomous Agents: A Closer Look at Why They Fail When Completing Tasks. *arXiv preprint arXiv:2508.13143* (2025). https://arxiv.org/abs/2508.13143

[43] Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. 2023. SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. *arXiv preprint arXiv:2303.08896* (2023).

[44] Ananya Mantravadi, Shivali Dalmia, Olga Pospelova, Abhishek Mukherji, Nand Dave, and Anudha Mittal. 2025. LegalWiz: A Multi-Agent Generation Framework for Contradiction Detection in Legal Documents. *arXiv preprint arXiv:2510.03418* (2025). Corrected authors (previously attributed to Zhang)..

[45] Dasha Metropolitansky and Jonathan Larson. 2025. VeriTrail: Closed-Domain Hallucination Detection with Traceability. *arXiv preprint arXiv:2505.21786* (2025).

[46] Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 12076–12100.

[47] Sonia Natalie Mitchell, Andrew Lahiff, Nathan Cummings, Jonathan Hollocombe, Bram Boskamp, Ryan Field, Dennis Reddyhoff, Kristian Zarebski, Antony Wilson, Bruno Viola, et al. 2022. FAIR Data Pipeline: provenance-driven data management for traceable scientific workflows. *Philosophical Transactions of the Royal Society A* 380, 2233 (2022), 20210300. doi:10.1098/rsta.2021.0300

[48] Luc Moreau and Paul Groth. 2013. *Provenance: An Introduction to PROV*. Morgan & Claypool Publishers.
[49] Luc Moreau, Paolo Missier, Khalid Belhajjame, Reza B'Far, James Cheney, Sam Coppens, Yolanda Gil, Paul Groth, Graham Klyne, Timothy Lebo, et al. 2013. *PROV-DM: The PROV Data Model*. W3C Recommendation. World Wide Web Consortium (W3C). https://www.w3.org/TR/prov-dm/
[50] Akshat Naik, Patrick Quinn, Guillermo Bosch, Emma Gouné, Francisco Javier Campos Zabala, Jason Ross Brown, and Edward James Young. 2025. Agent-Misalignment: Measuring the Propensity for Misaligned Behaviour in LLM-Based Agents. arXiv:2506.04018 [cs.AI] https://arxiv.org/abs/2506.04018
[51] Thomas Pasquier, Xueyuan Han, Mark Goldstein, Thomas Moyer, David Eyers, Margo Seltzer, and Jean Bacon. 2017. Practical Whole-System Provenance Capture. In *Proceedings of the 2017 Symposium on Cloud Computing (SoCC '17)*. 405–418.
[52] Luca Della Pellegrina, Ylber Haxhimusa, Raza Malik, et al. 2025. AI for scientific integrity: Detecting ethical breaches, errors, and fraud using generative AI tools and large language models. *Frontiers in Artificial Intelligence* 8 (2025), 1644098. doi:10.3389/frai.2025.1644098
[53] Karl Popper. 1959. *The Logic of Scientific Discovery*. Hutchinson, London.
[54] Paul Rasmussen and The Zep Team. 2025. Zep: A Temporal Knowledge Graph Architecture for Agent Memory. *arXiv preprint arXiv:2501.13956* (2025).
[55] Anjali Sharma, Christopher Lin, et al. 2025. ResearchRubrics: Auditing the Citation Integrity of Deep Research Agents. *arXiv preprint arXiv:2511.07685* (2025).
[56] Manasi Sharma, Chen Bo Calvin Zhang, Chaithanya Bandi, Clinton Wang, Ankit Aich, Huy Nghiem, Tahseen Rabbani, Ye Htet, Brian Jang, Sumana Basu, Aishwarya Balwani, Denis Peskoff, Marcos Ayestaran, Sean M. Hendryx, Brad Kenstler, and Bing Liu. 2025. ResearchRubrics: A Benchmark of Prompts and Rubrics For Evaluating Deep Research Agents. arXiv:2511.07685 [cs.AI] https://arxiv.org/abs/2511.07685
[57] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. 2024. AI models collapse when trained on recursively generated data. *Nature* 631 (2024), 755–759. doi:10.1038/s41586-024-07566-y
[58] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. AI models collapse when trained on recursively generated data. *Nature* 631, 8022 (01 Jul 2024), 755–759. doi:10.1038/s41586-024-07566-y
[59] Sameer Kumar Singh. 2026. State Drift in Language-Conditioned Autonomous Agents: A Failure Mode of Long-Horizon Reasoning. *Preprints* (January 2026). doi:10.20944/preprints202601.0910.v1
[60] Kevin Song, Anand Jayarajan, Yaoyao Ding, Qidong Su, Zhanda Zhu, Sihang Liu, and Gennady Pekhimenko. 2025. Aegis: Taxonomy and Optimizations for Overcoming Agent-Environment Failures in LLM Agents. arXiv:2508.19504 [cs.MA] https://arxiv.org/abs/2508.19504
[61] Renan Souza, Rafael Ferreira da Silva, et al. 2024. Workflow Provenance in the Computing Continuum for Responsible, Trustworthy, and Energy-Efficient AI. In *Proceedings of the 2024 IEEE International Conference on Cluster Computing (CLUSTER)*. IEEE.
[62] Renan Souza, Amal Gueroudji, Stephen DeWitt, Daniel Rosendo, Tirthankar Ghosal, Robert Ross, Prasanna Balaprakash, and Rafael Ferreira da Silva. 2025. PROV-AGENT: Unified Provenance for Tracking AI Agent Interactions in Agentic Workflows. In *2025 IEEE 21st International Conference on e-Science (e-Science)*. IEEE.
[63] Yoo Yeon Sung, Hannah Kim, and Dan Zhang. 2025. VeriLA: A Human-Centered Evaluation Framework for Interpretable Verification of LLM Agent Failures. arXiv:2503.12651 [cs.AI] https://arxiv.org/abs/2503.12651
[64] Nathan J Szymanski, Bernardus Rendy, Yuxing Fei, Rishi E Kumar, Tanjin He, David Milsted, Matthew J McDermott, Max Gallant, Ekin Dogus Cubuk, Amil Merchant, et al. 2023. An autonomous laboratory for the accelerated synthesis of novel materials. *Nature* 624, 7990 (2023), 86–91. doi:10.1038/s41586-023-06734-w
[65] SpecMAS Team. 2025. SpecMAS: A Multi-Agent System for Self-Verifying System Generation via Formal Model Checking. In *Advances in Neural Information Processing Systems (NeurIPS)*.
[66] Pranav Narayanan Venkit, Philippe Laban, Yilun Zhou, Kung-Hsiang Huang, Yixin Mao, and Chien-Sheng Wu. 2026. DeepTRACE: Auditing Deep Research AI Systems for Tracking Reliability Across Citations and Evidence. In *The Fourteenth International Conference on Learning Representations*. https://openreview.net/forum?id=QkaeTea16Y
[67] Andrew Wang, Sophia Hager, Adi Asija, Daniel Khashabi, and Nicholas Andrews. 2025. Hell or High Water: Evaluating Agentic Recovery from External Failures. In *Second Conference on Language Modeling*. https://openreview.net/forum?id=Zk224WPT42
[68] Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, et al. 2024. A Survey on Large Language Model based Autonomous Agents. *arXiv preprint arXiv:2308.11432* (2024).
[69] Yuxia Wang, Bela GIpp, and G Spanakis. 2024. Factcheck-GPT: End-to-End Fact-Checking with Large Language Models. *arXiv preprint arXiv:2311.09657* (2024).
[70] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, et al. 2022. Emergent Abilities of Large Language Models. *Transactions on Machine Learning Research* (2022). https://openreview.net/forum?id=yzkSU5zdwD
[71] Robert Wolfe, Isaac Slaughter, Bin Han, Bingbing Wen, Yiwei Yang, Lucas Rosenblatt, Bernease Herman, Eva Brown, Zening Qu, Nic Weber, and Bill Howe. 2024. Laboratory-Scale AI: Open-Weight Models are Competitive with ChatGPT Even in Low-Resource Settings. *arXiv preprint arXiv:2405.16820* (2024). https://arxiv.org/abs/2405.16820
[72] Boming Xia, Qinghua Lu, Liming Zhu, Zhenchang Xing, Dehai Zhao, and Hao Zhang. 2024. Evaluation-Driven Development and Operations of LLM Agents: A Process Model and Reference Architecture. In *arXiv preprint arXiv:2411.13768*.
[73] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P Xing, et al. 2024. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems* 36 (2024).
[74] Qinhao Zhou, Zihan Zhang, Xiang Xiang, Ke Wang, Yuchuan Wu, and Yongbin Li. 2024. Enhancing the General Agent Capabilities of Low-Parameter LLMs through Tuning and Multi-Branch Reasoning. *arXiv preprint arXiv:2403.19962* (2024). https://arxiv.org/abs/2403.19962
[75] Jialiang Zhu, Gongrui Zhang, Xiaolong Ma, Lin Xu, Miaosen Zhang, Ruiqi Yang, Song Wang, Kai Qiu, Zhirong Wu, Qi Dai, Ruichun Ma, Bei Liu, Yifan Yang, Chong Luo, Zhengyuan Yang, Linjie Li, Lijuan Wang, Weizhu Chen, Xin Geng, and Baining Guo. 2025. RE-TRAC: REcursive TRAjectory Compression for Deep Search Agents. *arXiv preprint arXiv:2602.02486* (2025).
[76] Minjun Zhu, Qiujie Xie, Yixuan Weng, Jian Wu, Zhen Lin, Linyi Yang, and Yue Zhang. 2025. AI Scientists Fail Without Strong Implementation Capability. *arXiv preprint arXiv:2506.01372* (2025).